

Presents

CODERUSH 1.0

A National level 24 hr hackathon

Team Name : **INSIGHT MINERS**

Team Leader Name : Pritik Kumar Mohanty

Problem Statement : **AI-Powered Solar Panel Dust Detection & Automated Cleaning System**



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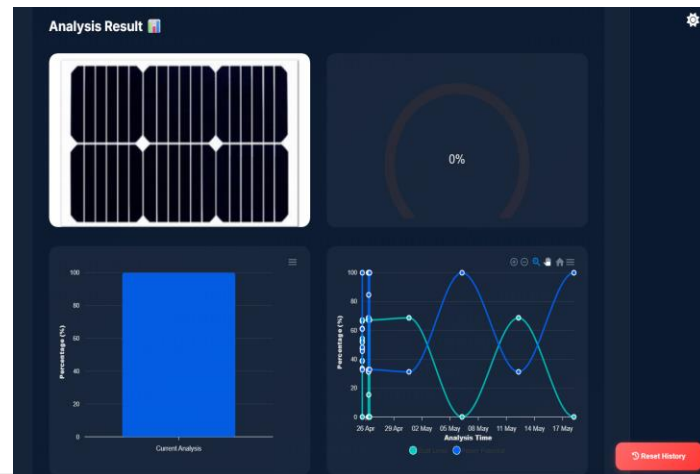
LinkedIn: <https://www.linkedin.com/in/sangram-dalave-840a58245/>



Brief about the Idea:

SOLARENSE is an AI-powered autonomous system that revolutionizes solar panel maintenance through intelligent dust detection and automated cleaning.

- **The Big Problem:** India loses ₹15,000 crores annually due to dust-reduced solar efficiency (20-30% power loss). With 75 GW installed capacity growing to 500 GW by 2030, this problem will cost ₹1 lakh crore in lost energy.
- **Our Solution:** Patent-pending **VGG16-SVM** hybrid AI algorithm detects dust in real-time, triggers water-efficient automated cleaning, and provides predictive maintenance - all at 70% lower cost than existing robotic solutions.
- **Market Impact:** Targeting ₹1,260 crore solar cleaning market with our ₹25,000-50,000 system vs ₹50+ lakh competitors.



OUR SOLUTION



AI-Powered Dust Detection

Hybrid VGG16–SVM model identifies dust in real-time with ~73–75% accuracy.



Automated Cleaning System

Robot cleaners triggered instantly by AI → timely, efficient, and water-saving.



IoT + Edge Integration

Continuous monitoring, predictive maintenance, and <30ms response time.



Sustainable & Affordable

Uses little/no water, reduces O&M costs by **40%**, boosts energy output by **25–30%**.



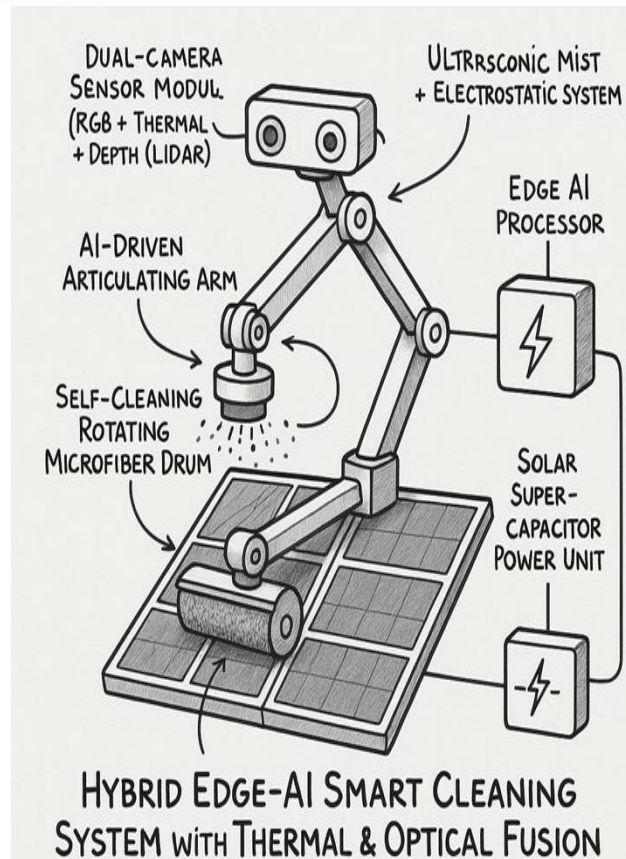
Patented AI detection

Ensures a unique and high-accuracy cleaning system not offered by others

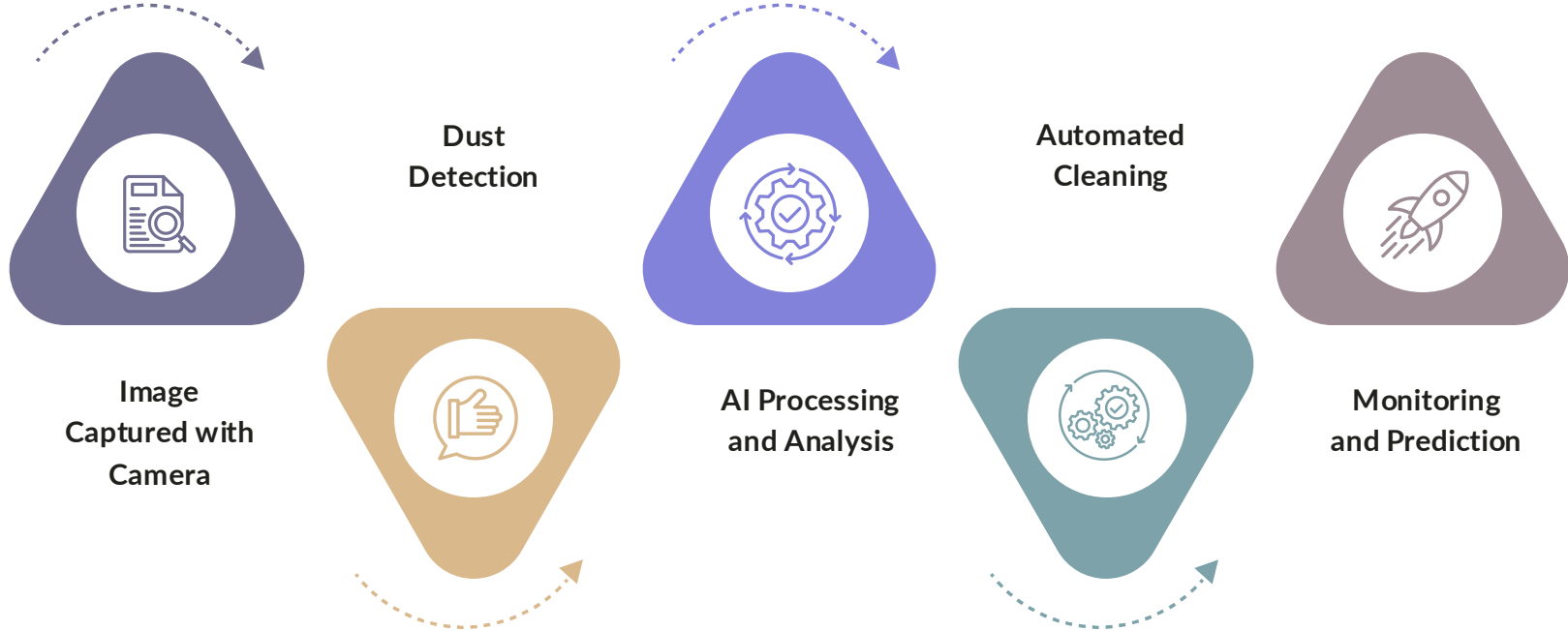
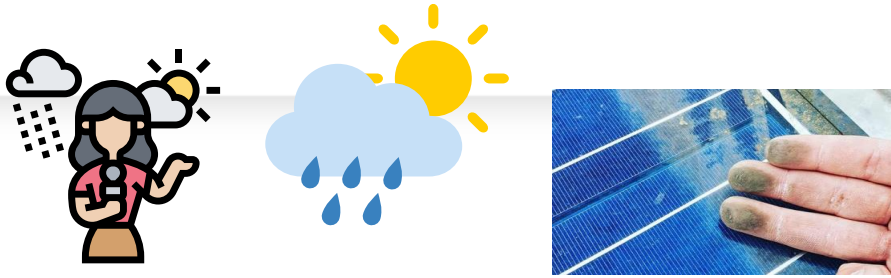
List of features offered by the solution

CORE FEATURES:

1. AI Dust Detection Engine: **Real-time computer vision** analysis
2. Automated Cleaning System: **Motorized brush + ultrasonic mist spray**
3. Predictive Maintenance: **Weather-based cleaning** schedule optimization
4. IoT Dashboard: **Remote monitoring** and control via mobile/web app
5. Water Conservation Mode: **90% reduction in water usage**
6. Emergency Response: Instant cleaning during dust storms
7. Performance Analytics: ROI tracking and efficiency reports
8. Modular Design: **Expandable** for different solar installation sizes
9. Edge Computing: Local processing for <30-second response time

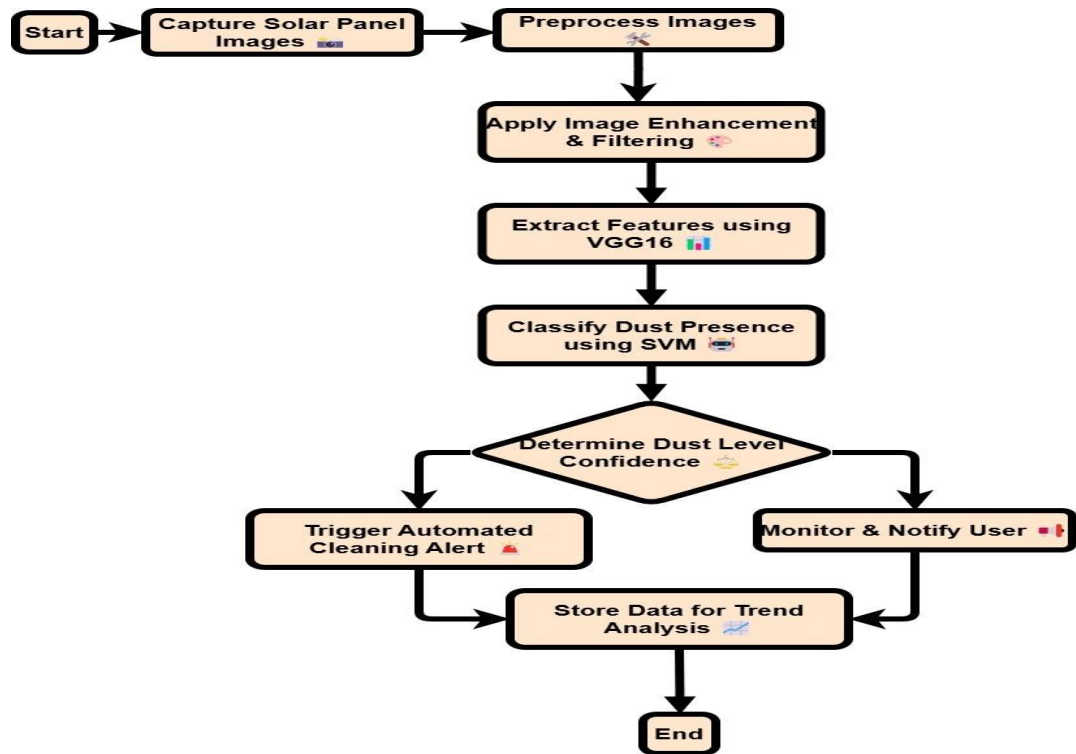
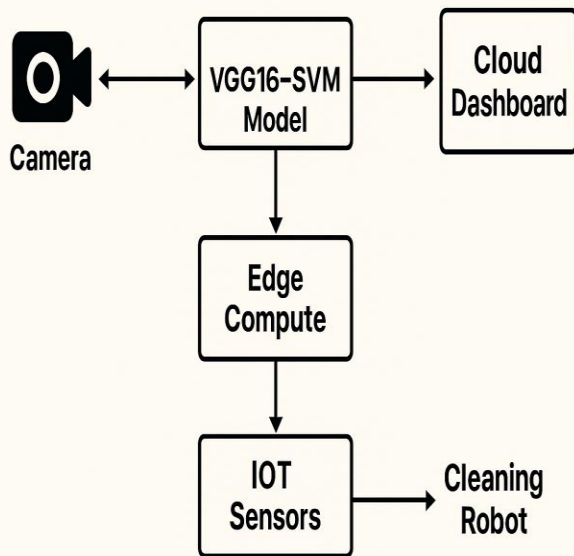


Methodology ???

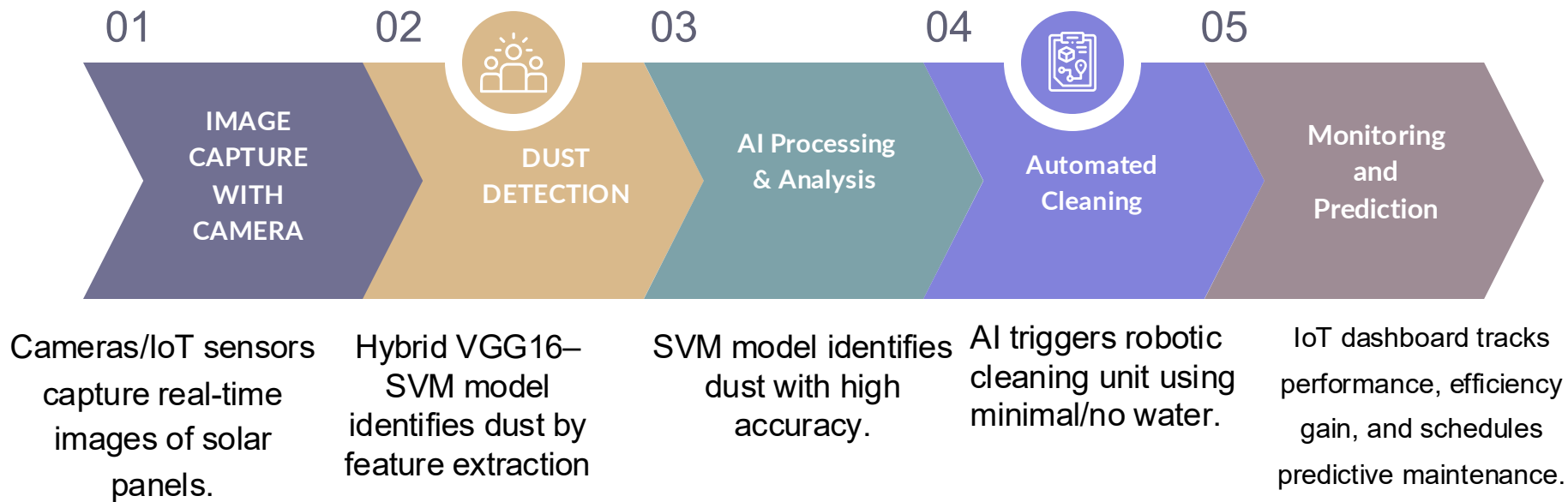


Process flow diagram or Use-case diagram

SOLAR PANEL AI DUST DETECTION

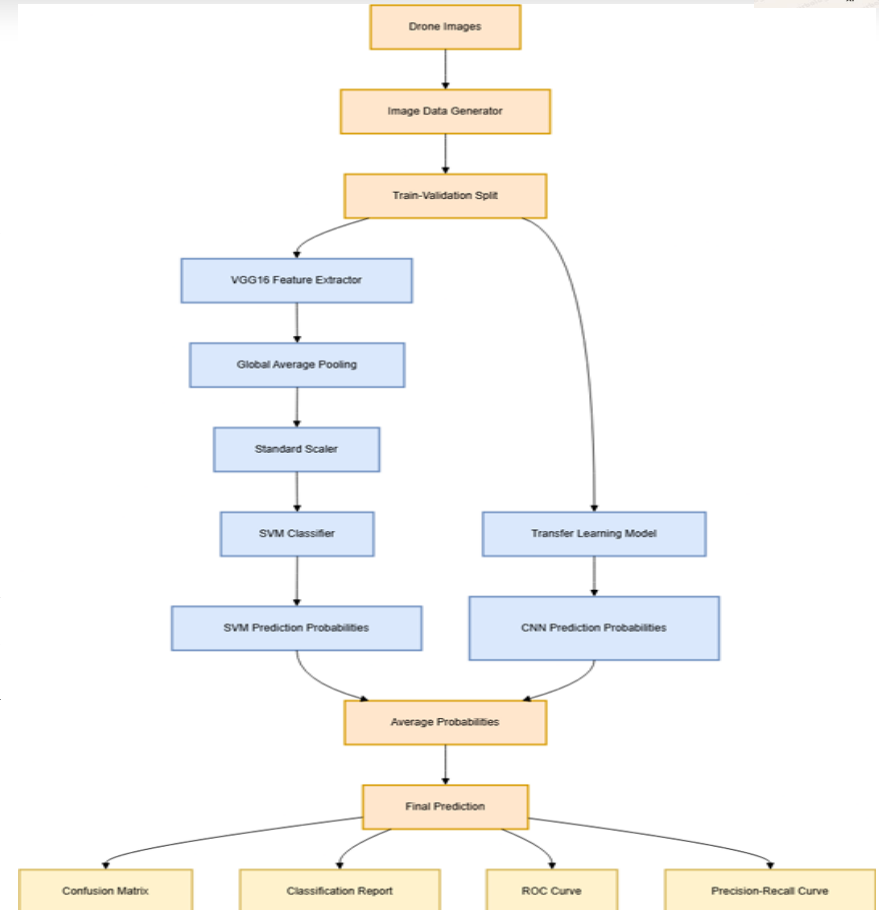


MOCK DIAGRAM



Architecture diagram of the proposed solution:

This architecture diagram illustrates the process of a system to detect dust on a solar panel with drone imagery and machine learning. It outlines the process from drone images, train-validation with train-validation splits in order to get the data ready for training models. The training data is then passed through a VGG16 feature extractor to get meaningful features from the training data, global average pooling, and standardization by using a Standard Scaler. These features are input into an SVM classifier to produce SVM prediction probabilities.



TECHNOLOGY STACK

Frontend

- **HTML, CSS, JavaScript:** Web interface and user experience
- **Responsive design for phones & tablets**
- **OpenCV:** Real-time image processing



APIs & AI Engine

- **Open Weather Map / Tomorrow.io** for temperature, precipitation, UV index
- **Weather API Integration:** Data-driven predictive maintenance



Backend & Database

- **Node.js, MongoDB:** Backend server and data storage, React.js: Mobile app and web dashboard development
- **TensorFlow, PyTorch:** Deep learning and model training
- **VGG16:** CNN based model for working image data
- **Machine Learning: SVM (Support Vector Classifier)**



IOT HARDWARE USED

- **AWS IoT Core:** Cloud device management and data streaming
- **Raspberry Pi 4B, ESP32:** Edge computing and IoT connectivity
- **Camera Drone** for capturing images of solar panels

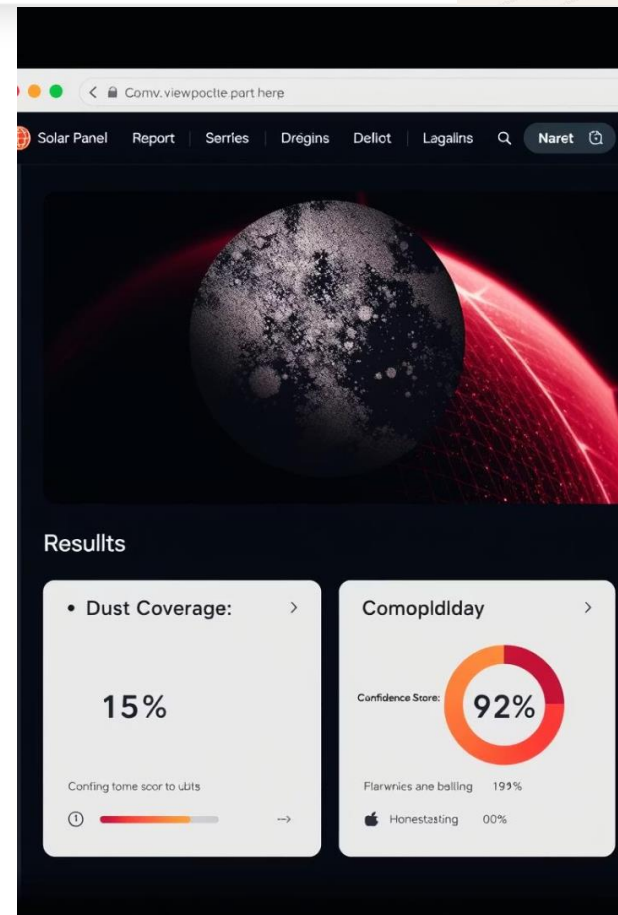


Estimated Implementation Cost:

- **Proposed System (SOLARENSE):** ₹25,000 - ₹40,000 per unit
- **Existing Robotic Solutions:** ₹30+ lakh per unit
- **Manual Cleaning (Recurring):** ₹3–5 lakh/year for large solar farms

Key Advantage:

- **~70% cheaper** than robotic cleaners
- **Payback period < 2 years** through higher energy output & lower O&M costs





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THANK YOU